Tanvir Mahmud Mahim

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Education

Dec 2022 / Jan 2019 — **B.Sc. in Electrical and Electronic Engineering (EEE), BRAC University**

- CGPA: **3.77/4.00** | Average Marks: **86.20%** | Academic Honors: **High Distinction**
- Certification: World Education Services (WES) Certified, Equivalent to a US Bachelor's Degree
- Supervisor: Dr. A.H.M.A. Rahim [Retired (Dec 2024)]
- Thesis: BRACU-DSpace-2023

Honors and Awards

Quality Journal Publication Award for Scopus-indexed Q1 journal publications (2025, 2026)
VC's and Dean's List Award for academic excellence at undergraduate study (2020-2022)
University Merit Scholarship for academic excellence at undergraduate study (2020-2022)

Research Interests

- Quantum Optics and Photonics
- 2D Quantum Materials
- Wide Bandgap (WBG) Semiconductor Devices
- Quantum Physics
- Electronic Design Automation (EDA)
- Machine Learning (ML)

Research Experience

Present / Jan 2025 — **Squeezed-Light Microcombs, Diamond Spin Qubits and Sensing, Photonic ML**
Supervisor: Dr. A.S.M. Mohsin, **BRACU, EEE**

- Developed a numerical model of a photonic-molecule 4H-SiC soliton-crystal microcomb enabling squeezing extraction beyond 3 dB limit [Code] [Optics Express 2026]
- Modeled diamond spin qubits for fast readout and quantum sensing, by theory and simulation: Over-coupled diamond nanobeam cavity on thin-film lithium niobate for single-shot readout of tin-vacancy spins, 98.8% fidelity in 0.1 μs [Code] [under review]; NV spin sensing proposed to identify which trapped vortices cause microwave loss in tantalum qubit films [Code] [under review]
- Developed a closed-loop ML framework for quantum photonic instrumentation, in collaboration with the University of Memphis, USA: Closed-loop, event-driven machine learning for autonomous triage of single-photon emitters [Code] [ACS Photonics 2026]
- Developed learned generative process twins for yield-aware inverse design of multilayer photonic sensor filters [Code] [IEEE Sensors Journal 2026]

July 2026 / July 2024 — **GaN & 2D Semiconductor Devices and Physics-Guided Design**
Supervisor: Dr. M. Mosaddequr Rahman, **BRACU, EEE**

- Traced the origin of the conflicting hole masses in the polarization-induced GaN/AlN 2D hole gas with a self-consistent six-band k·p analysis [Code] [Phys. Rev. B 2026]
- Showed when the channel drops out of the memory window of $CuInP_2S_6$-gated WSe_2 ferroelectric transistors, and how strain and interface traps act [Code] [Mater. Today Commun. 2026]
- Developed a fabrication-aware, physics-informed adjoint framework with RL for inverse design of CMUTs and photonic sensors. [Code], [under review], [IEEE Sensors Journal 2025]
- Developed physics-guided circuit design frameworks, in collaboration with Dr. Nadim Chowdhury, BUET: device-physics design rules for wide-temperature e-mode GaN-on-SOI integrated circuits, found by sizing a fractional-N PLL with transistor-anchored machine learning [Code] [under review]; graph RL with an adaptive fuzzy reward for transferable analog transistor sizing, with GlobalFoundries, Inc., USA [Code] [under review]

June 2024 / May 2023 — **Learning-Based Grid Control and Bifacial Photovoltaics**
Supervisor: Dr. A.H.M.A. Rahim, **BRACU, EEE** [Retired (Dec 2024)]

- Developed RL-based adaptive fuzzy inference to help DFIG wind turbines withstand extreme low voltages [JESTECH (Elsevier) 2026]
- Developed an RL-driven fuzzy-attention inference controller for microgrids. [IEEE TFS 2025]
- Developed a novel bifacial module [IEEE TENSYMP 2023], [IEEE JPV 2024]

Selected Publications (Find all here:)

2026 **T. M. Mahim**, M. M. Rahman and A. S. M. Mohsin, "Overcoming the 3 dB Squeezing Extraction Limit in Silicon Carbide Microcombs with a Photonic Molecule", ***Optics Express***, 34(18), 34822 (2026). [Paper]

2026 **T. M. Mahim**, M. N. Islam, M. M. Rahman and A. S. M. Mohsin, "Closed-Loop, Event-Driven Machine Learning for Autonomous Triage of Single-Photon Emitters", ***ACS Photonics***, published online (2026). [Paper]

2026 **T. M. Mahim**, A. S. M. Mohsin and M. M. Rahman, "Origin of the Conflicting Hole Masses in the GaN/AlN Two-Dimensional Hole Gas", ***Physical Review B*** (Letter), 114, L291304 (2026). [Paper]

2026 **T. M. Mahim** and M. M. Rahman, "Channel decoupling of the memory window in $CuInP_2S_6$-gated two-dimensional ferroelectric transistors: conditions, limits, and implications for interface processing", ***Materials Today Communications***, 57, 116333 (2026). [Paper]

2025 **T. M. Mahim**, M. M. Rahman and A. H. M. A. Rahim, "Hierarchical Inverse Design Framework for Unit-Cell CMUTs With Attentive Gated Recurrent And Fully Connected Dense Layers", ***IEEE Sensors Journal***, 25(13), 26104-26116 (2025). [Paper]

2025 **T. M. Mahim**, A. H. M. A. Rahim and M. M. Rahman, "Adaptive Fuzzy Attention Inference to Control A Microgrid Under Extreme Fault On Grid Bus", ***IEEE Transactions on Fuzzy Systems***, 33(6), 1815-1824 (2025). [Paper]

2024 **T. M. Mahim**, A. H. M. A. Rahim and M. M. Rahman, "Weather Responsive Multidimensional Photovoltaic Efficiency Model for Simulation of Custom-Built Bifacial Panel", ***IEEE Journal of Photovoltaics***, 14(5), 848-860 (2024). [Paper]

Open-Source Research Software (TaN-MM-Org · PyPI)

ramansep	Strain and doping from 2D Raman maps	**kpenvelope**	Six-band k·p solver for nitride layers
sqzcomb	Detected squeezing from Kerr microcombs	**absnoise**	Andreev-level noise in Josephson detectors
cavsqueeze	Cavity spin squeezing of spin ensembles	**sparq-triage**	Automated single-photon emitter screening
vacspin	Diamond color-center spin-photon interfaces	**hamop**	One tight-binding model, every property
fabtwin	Learned process twins for yield design	**fracpll**	Fractional-N PLL design from measured parts
lockkernel	Exact synchronization transitions and fits	**labplan**	Measurement planning and calibration

</> Technical Skills

TCAD & Device Simulation	*DEVSIM · Synopsys Sentaurus TCAD*
Multiphysics & Electromagnetics	*COMSOL Multiphysics · Ansys HFSS*
First-Principles & Atomistic	*Density functional theory (Quantum ESPRESSO) · Wannierization (Wannier90) · Molecular dynamics (LAMMPS) · Micromagnetics (MuMax^3)*
Circuit, Layout & EDA	*Cadence Virtuoso · Keysight ADS · Xyce · ngspice · OpenVAF · GDSFactory*
Machine Learning	*PyTorch · JAX, Optax, Flax · TensorFlow & Keras · scikit-learn*

Professional Experience

Present – June 2025 — **Adjunct Lecturer, Department of EEE, BRACU**
- Courses: Semiconductor Device Physics, Electrical Circuits II (Analog Design), Numerical Techniques and Computer Programming in C

Present – June 2023 — **Research Assistant, Department of EEE, BRACU**
- Collaborated with industries and international universities; assisted undergraduate capstone projects

June 2026 – Dec 2025 — **Consultant, GIXSystems.ai, USA**
- Technology development of a 100 V GaN-on-Si power HEMT discrete device (tape-out with X-FAB, Erfurt)

May 2023 – Jan 2022 — **Teaching Assistant, Department of EEE, BRACU**
- Courses: DC & Magnetic Circuits, Energy Conversion I and Signals & Systems

Synergistic and Leadership Activities

Present – Jan 2021 — **Student Member** | **IEEE** · Electron Devices Society (EDS) | Power and Energy Society (PES)

Nov 2023 – Feb 2023 — **Aspire Institute** | **Harvard University, USA** · Aspire Leaders Program 2023

Referees

Dr. Md. Mosaddequr Rahman · Professor & Chairperson, Dept. of EEE, BRAC University, Bangladesh · mosaddeq@bracu.ac.bd

Dr. Abu S. M. Mohsin · Associate Professor, Department of EEE, BRAC University, Bangladesh · asm.mohsin@bracu.ac.bd

Dr. A.S. Nazmul Huda · Associate Professor, Department of EEE, BRAC University, Bangladesh · nazmul.huda@bracu.ac.bd